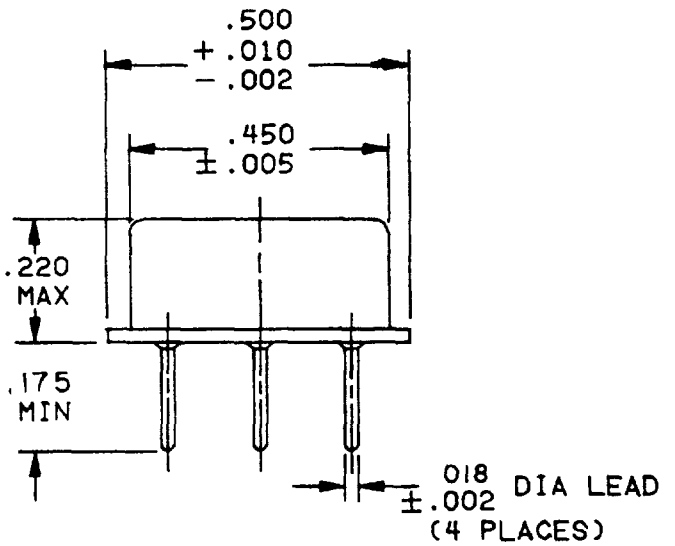
MILITARY SPECIFICATION SHEET

MIXER STAGES, RADIO FREQUENCY, DOUBLE BALANCED, TO CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the mixers described herein shall consist of this specification and the latest issue of MIL-M-28837.



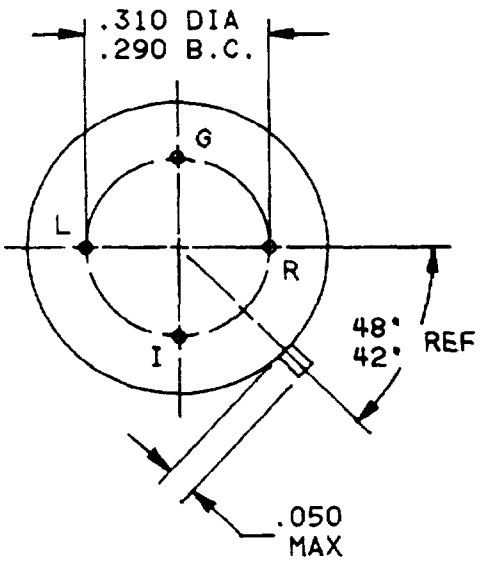


FIGURE 1. Outline drawing for mixers P/N M28837/9-01 through -04.

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INCHES	MM
.002	0.05
.005	0.13
.010	0.25
.017	0.43
.050	1.27
.200	5.08
.220	5.59
.310	7.87
.450	11.43
.500	12.70

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Outline drawing for mixers P/N M28837/9-01 through -04 - Continued.

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REQUIREMENTS.

Design and construction

Dimensions and configuration: See figure 1.

Electrical characteristics:

Operating frequency range: See table I.

LO drive power. See table I.

Conversion loss (max). See table I.

Noise figure (SSB) See table I.

Isolation (minimum). See table I.

Maximum input power See table I.

Conversion compression. See table I.

Desensitization (max) See table I.

DC relative polarity See table I.

VSWR. See table I

Third order two tone intermodulation See table I.

Impedance 50 ohms.

Physical and environmental characteristics:

Weight: .07 ounce (2 grams) maximum.

Temperature.

Operating. -54°C to +100°C.

Nonoperating. -65°C to +100°C

Shock Test condition C, MIL-STD-202.

Life. 100,000 hours minimum.

Seal Hermetic.

Part number. M28837/9- (dash number from table I).

Substitutability data: See table II.

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TABLE I Electrical characteristics

Dash no	Operating frequency range (MHz)	LO input drive power in dBm 1/	Maximum conversion loss (SSB) dB (MHz)	Noise figure (SSB) dB (MHz)	Isolation Minimum LO-RF dB	LO-IF dB	RF-IF dB	Frequency (MHz)
01N 01S	RF 10-1000 LO 10-1000 IF DC-1200	minimum -3 TV 0 maximum +5	8.5 at f_L and f_R 10-1000 f_I DC-1200	Within 1 dB of conversion loss	40	35	30	10-50
					20	25	15	50-750
					15	17	10	750-1000
02N 02S	RF 10-1000 LO 10-1000 IF DC-600	minimum +17 TV +20 maximum +24	7.5 at f_L and f_R 10-300 f_I 10-200	7.5 at f_L and f_R 10-300 f_I 10-200	35	35		10-100
					25	25		100-400
			9.0 at f_L and f_R 10-300 f_I 1-600	9.0 at f_L and f_R 10-300 f_I 2-600	18	18		400-1000
							12	10-1000
			10 at f_L and f_R 10-1000 f_I 1-500	10 at f_L and f_R 10-1000 f_I 2-500				
03N 03S	RF 10-1500 LO 10-1500 IF DC-800	minimum +7 TV +7 maximum +13	7.2 at f_R 20-600 f_L 10-800 f_I 1-200	7.2 at f_R 20-600 f_L 10-800 f_I 1-200	35	30		10-500
					28	20		500-1200
			9.0 at f_L and f_R 10-1500 f_I 1-800	9.0 at f_L and f_R 10-1500 f_I 1-800	25	18		1200-1500
04N 04S	RF 10-1600 LO 10-1600 IF DC-800	minimum +11 TV +13 maximum +17	7.5 at f_L 10-800 f_R 20-600 f_I 0.4-200	7.5 at f_L 10-800 f_R 20-600 f_I 0.4-200	35	28		10-500
					28	20		500-1000
			8.5 at f_L and f_R 10-1600 f_I 0.4-800	8.5 at f_L and f_R 10-1600 f_I 0.4-800	25	15		1000-1600

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash no.	RF & LO 2/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 3/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
01N 01S	200 mW	1.0 dB @ P_R -8 dBm	1.0 dB @ P_{R2} -10 dBm	Positive	2.0 1	---	2.0·1	+7 4/	f_{LO}: 600 MHz f_{R1}: 500 MHz @ -20 dBm f_{R2}: 505 MHz @ -20 dBm
02N 02S	500 mW	1.0 dB @ P_R +20 dBm	1.0 dB @ P_{R2} +18 dBm	Negative	2.5·1	2 0 1	3.5·1	+25 4/ +23	f_{LO}: 15 MHz @ +23 dBm f_{R1}: 10 MHz @ 0 dBm f_{R2}: 11 MHz @ 0 dBm f_{LO}: 400 MHz @ +23 dBm f_{R1}: 1000 MHz @ 0 dBm f_{R2}: 900 MHz @ 0 dBm
03N 03S	200 mW	1.0 dB @ P_R 0 dBm	1.0 dB @ P_{R2} -2 dBm	Positive	2.1 1	2.8 1	3.0 1	+12 4/ +10	f_{LO} 15 MHz f_{R1}: 10 MHz @ -10 dBm f_{R2}: 11 MHz @ -10 dBm f_{LO}: 1500 MHz f_{R1} 1400 MHz @ -10 dBm f_{R2} 1410 MHz @ -10 dBm
04N 04S	200 mW	1.0 dB @ P_R +7 dBm	1.0 dB @ P_{R2} +5 dBm	Positive	2.8 1	3 0 1	3.0·1	+18	f_{LO}: 1150 MHz f_{R1} 500 MHz @ -5 dBm f_{R2} 505 MHz @ -5 dBm

1/ Unless otherwise specified, TV is the test value of the LO power for the electrical characteristics.

2/ These values are for +25°C and are derated linearly to +100°C.

3/ With two in-phase signals applied to the LO and RF ports.

4/ Input intercept points.

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Substitutability data. The mixers covered by this specification sheet are substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN
01	12813	645018-1	---
01	05375	DBM-191	---
02	14482	M2E	---
03	96214	807714-1	---
03	14482	M2A	---
04	05869	729032-1	---
04	05869	729032-2	---
04	14482	M2B	---

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC
Air Force - 14, 19

Agent.
DLA - ES

Preparing activity:
Navy - EC

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

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c. Reason/Rationale for Recommendation

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